

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{8}{243} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{81} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{1215} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_d^{\mathbf{P}}] \right. \\
& \quad \delta_{i1i2} \delta_{i3i4} - \frac{8}{81} \sum_{\mathbf{p}} g_1^4 LF_{3,0}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{27} \sum_{\mathbf{p}} g_1^4 LF_{4,-1}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{32}{405} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} \sum_{\mathbf{l}} g_1^4 LF_{3,0}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{5}{54} \sum_{\mathbf{l}} g_1^4 LF_{4,-1}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{405} \sum_{\mathbf{l}} g_1^4 LF_{5,-2}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{243} \sum_{\mathbf{q}} g_1^4 LF_{3,0}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{162} \sum_{\mathbf{q}} g_1^4 LF_{4,-1}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{16}{1215} \sum_{\mathbf{q}} g_1^4 LF_{5,-2}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{243} \sum_{\mathbf{u}} g_1^4 LF_{3,0}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{20}{81} \sum_{\mathbf{u}} g_1^4 LF_{4,-1}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{128}{1215} \sum_{\mathbf{u}} g_1^4 LF_{5,-2}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{81} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{27} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{32}{405} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{243} g_1^4 LF_{2,1,0}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{2,2,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{243} g_1^4 LF_{3,1,-1}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{4,1,-2}[m_1, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{1}{243} g_1^4 LF_{2,1,0}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{2,2,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{243} g_1^4 LF_{3,1,-1}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{4,1,-2}[m_1, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{243} g_1^4 LF_{2,1,0}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{2,2,-1}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{243} g_1^4 LF_{3,1,-1}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{4,1,-2}[m_1, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{243} g_1^4 LF_{2,1,0}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{2,2,-1}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{243} g_1^4 LF_{3,1,-1}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{4,1,-2}[m_1, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i3}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i4}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_u^{i1}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{4}{81} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_u^{i2}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{243} g_1^4 LF_{2,1,0}[m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{3,1,-1}[m_d^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{2}{243} g_1^4 LF_{2,1,0}[m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{243} g_1^4 LF_{3,1,-1}[m_d^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{27} g_1^2 (-2 \overline{y}_d^{pi3} y_d^{pi4} \delta_{i1i2} + \overline{y}_u^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,1,0}[m_q^{\mathbf{P}}, \tilde{\mu}] + \\
& \quad \frac{1}{54} g_1^2 (2 \overline{y}_d^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y}_u^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,2,-1}[m_q^{\mathbf{P}}, \tilde{\mu}] + \\
& \quad \frac{1}{54} g_1^2 (2 \overline{y}_d^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y}_u^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{3,1,-1}[m_q^{\mathbf{P}}, \tilde{\mu}] + \\
& \quad \frac{8}{243} g_1^4 LF_{2,1,0}[m_u^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{3,1,-1}[m_u^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_u^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_u^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{243} g_1^4 LF_{2,1,0}[m_u^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{4}{243} g_1^4 LF_{3,1,-1}[m_u^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{8}{81} g_1^2 g_3^2 LF_{2,1,0}[m_u^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^2 g_3^2 LF_{3,1,-1}[m_u^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{54} g_1^2 (2 \overline{y}_d^{pi3} y_d^{pi4} \delta_{i1i2} - \overline{y}_u^{pi1} y_u^{pi2} \delta_{i3i4}) LF_{2,1,0}[\tilde{\mu}, m_q^{\mathbf{P}}] + \\
& \quad \frac{1}{54} g_1^2 (14 \overline{y}_d^{pi3} y_d^{pi4} \delta_{i1i2} + 11 \overline{y}_u^{pi1} y_u^{pi2} \delta_{i3$$